

Financial and Operational Analysis of the Mexico City Trolleybus Network: Focus on Line 2 (Pantitlán-Chapultepec) for Infrastructure Integration Models

The transformation of the mobility paradigm in Mexico City has positioned the Electric Transport Service (STE) as the guiding axis of the transition towards a zero-emission fleet. Trolleybus Line 2, historically known as the Eje 2-2A Sur Zero Emissions Corridor, represents a fundamental case study for understanding the economics of mass urban transportation. This technical report breaks down the financial, operational, and technical indicators necessary for the construction of mathematical optimization models, integrating data from the Public Account, transparency reports, and engineering specifications of the rolling stock updated for the 2023-2025 period.

Institutional Context and Budgetary Framework of the Electric Transport Service

The analysis of operation and maintenance costs (OPEX) at STE must start from the public expenditure structure defined by the Secretariat of Administration and Finance of Mexico City. For the 2023 fiscal year and the 2024-2025 projections, the body has operated under a consolidation scheme of the Integrated Mobility Network (MI), which implies a harmonization not only of the image and payment but also of shared cost structures.

In 2023, Mexico City registered an inflation rate of 4.7%, a factor that directly affected the cost of imported inputs and salary negotiations for technical personnel. The 2023 Public Account reveals that the general Operating Expenditure of the public administration experienced a real variation of -9.6% compared to 2022, reflecting an austerity policy that, however, prioritized strategic sectors such as transport. The transport sector, as a whole, managed an exercised budget of 182,774.8 million pesos at the central level, with significant variations allocated to the modernization of electric systems.

The Electric Transport Service, as a decentralized body, manages its resources through specific budgetary programs. Program E042, titled "Operation and Maintenance of Mass, Concessioned, and Alternative Public Transport," is the financial backbone for Line 2. The main goal of this program is to guarantee progress in passenger transportation and the execution of preventive maintenance actions for the correct functioning of the fleet.

Operation and Maintenance Cost Structure (OPEX)

The cost of operation per kilometer is a composite indicator that integrates electrical energy consumption, personnel services, and preventive and corrective maintenance. For 2024, the STECDMX Draft Expenditure Budget defines clear allocations that allow the inference of Line 2's operating cost.

In the category of Personnel Services (Chapter 1000), Program E042 has an allocation of 834,046,847.00 pesos for 2024. This amount covers the salaries of trolleybus operators, as well as the technical personnel in charge of the electrical infrastructure and rolling stock. It is relevant to note that Line 2 operates with 12-meter units and has recently integrated 18-meter articulated units to absorb demand derived from partial closures on other Metro lines, which increases the intensity of personnel use per kilometer traveled.

The Materials and Supplies chapter (Chapter 2000) for the same program amounts to 72,109,177.00 pesos. This includes critical spare parts, tires, and carbon brushes for the trolleybus poles. Finally, the General Services chapter (Chapter 3000), which encompasses the payment of electrical energy to the Federal Electricity Commission (CFE) and external maintenance services, has 413,525,562.00 pesos available.

Expense Category (Program E042)	2024 Budget Allocation (MXN)	Operational Relevance
Personnel Services	834,046,847.00	Salaries, benefits, and social security for operational personnel.
Materials and Supplies	72,109,177.00	Acquisition of spare parts, lubricants, and cleaning materials.
General Services	413,525,562.00	Electrical energy (CFE), insurance, and contracted maintenance.
Total Current Expenditure	1,319,681,586.00	Total budget for the annual operation of the STE systems.

Note: The data corresponds to Program E042 which integrates Trolleybus, Light Rail, and Cablebús.

For a mathematical model, the OPEX per kilometer is calculated by dividing the total expenditure assigned to the trolleybus modality by the annual mileage traveled. Line 2 has a length of 18.0 km per direction, totaling 36.0 km per complete round trip. With an approximate departure frequency of every 8 minutes, the line generates a higher density of kilometers traveled than peripheral corridors. International benchmarking research suggests that the maintenance cost per kilometer for articulated trolleybus systems is around 0.123 USD (approximately 2.15 MXN at the average exchange rate of 2024), which is consistent with the cost structure observed at STE.

Analysis of Technical Complexity and Wear Penalty

An infrastructure integration model requires quantifying how route geometry affects component lifecycle. Line 2, although running mostly along leveled avenues such as Eje 2 Sur, presents technical challenges at its terminals and at crossings with elevated structures and underpasses.

Impact of Curves and Track Geometry on Rolling Stock

Tire wear is one of the most sensitive items in fleet maintenance. In the context of mass public transportation, the useful life of a tire is significantly reduced in sections with tight curves due to lateral forces and constant friction. An average transport vehicle in Mexico travels about 15,000 km annually, but a Line 2 trolleybus can reach this mileage in less than three months of intensive operation.

Trolleybus maintenance has an estimated premium of 10% compared to diesel buses with similar characteristics, mainly due to the complexity of the electric traction systems and the interaction with the catenary. However, this operational penalty is compensated by a vehicle life that reaches 20 years, compared to 10 to 12 years for internal combustion units.

Wear of the Electrical Contact Infrastructure

The technical diagnosis of the catenary is crucial to avoid operational failures. In critical sections, problems of abnormal contact wire wear caused by lack of mechanical tension have been identified. The elastic catenary requires precise adjustment; insufficient tension in the carrier or the copper wire generates "electrical arcing" (hot spots) as the trolleys pass, which prematurely carbonizes both the overhead conductor and the vehicle's carbon brushes.

For modeling purposes, the penalty for technical complexity can be expressed as a factor of increase in preventive maintenance cost. In sections with small radius curves or steep slopes (such as access to underpasses), the consumption of carbons and contact wire wear can increase between 15% and 25% compared to level straight sections.

Component	Cause of Wear	Implication in OPEX
Contact Wire (Copper)	Electrical arcing due to lack of tension or leveling.	Need for total change of the carrier in 1500m sections.

Component	Cause of Wear	Implication in OPEX
Carbon Brushes	Mechanical friction and erosion by electrical sparks.	Frequent replacement in high-frequency passage areas.
Tires	Maneuvers in tight curves and dense terminals.	Accelerated replacement cycles due to intensive mileage.

Elevated Infrastructure and its Effect on Maintenance Costs

The interaction of the trolleybus with elevated structures, either through underpasses or in the specific case of the Elevated Trolleybus (Line 10), introduces additional cost variables that must be differentiated from the surface operation of Line 2.

Maintenance of Elevated Stations and Tracks

Stations that interact with elevated structures or underpasses require more expensive life support and accessibility systems. For example, the Elevated Trolleybus in Ermita Iztapalapa operates with a fare of 7 pesos, higher than the 4 pesos of the surface Line 2, partly reflecting the higher maintenance cost of the viaduct infrastructure.

The maintenance of an elevated viaduct infrastructure for trolleybuses has an estimated cost of 0.5% of the initial investment value annually. In absolute terms, the CAPEX of a trolleybus infrastructure with partial catenary on a viaduct can reach 2.7 million USD per kilometer. For Line 2, although it is surface level, sections crossing vehicular bridges or urban viaducts require periodic structural inspections and a more complex catenary maintenance logistics, as the use of tower trucks is limited by the height and surrounding vehicular flow.

Cost Comparison: Surface vs. Elevated

STE's experience with the Cablebús system provides a reference point for the maintenance costs of non-surface transport systems. It is estimated that the maintenance of elevated transport lines requires an annual budget of approximately 50 million pesos per line, covering the inspection of cabins, motors, and support structures. Although the trolleybus does not require the same complexity of traction cables, the maintenance of the electrical substations that feed elevated sections must be more rigorous to avoid interruptions in areas of difficult evacuation.

Infrastructure Attribute	Surface Trolleybus (L2)	Elevated Trolleybus (L10)	Impact on the Model
Public Fare	\$4.00 MXN	\$7.00 MXN	Reflects the subsidy and the recovery cost.
Infrastructure CAPEX	Low-Medium	High (~2.7 M USD/km)	Initial investment amortized over 30-50 years.
Track Maintenance	Standard Pavement	Structural (Viaduct)	0.5% annual increase in CAPEX.
Accessibility	Street level	Elevators and stairs	Higher general services cost (electricity, cleaning).

Analysis of Energy Consumption and Yutong Fleet Efficiency

Energy consumption is the most dynamic component of OPEX. The modernization of the Line 2 fleet with units from the manufacturer Yutong has allowed for a substantial improvement in energy efficiency and operational flexibility through the use of batteries.

Technical Specifications and Consumption per Kilometer

Standard 12-meter Yutong units (Series 20000, 22000, 23000) operate with Lithium Iron Phosphate (LFP) batteries with a capacity of 127 kWh. These units are designed for a maximum speed of 100 km/h and can transport up to 85 passengers. The most relevant technical characteristic for the integration model is their autonomy of at least 70 km using only the secondary battery system.

In terms of actual consumption, operation data in Mexico City for electric buses of similar dimensions (such as those used in Metrobús lines) indicate an average consumption of 1.0 kWh per kilometer traveled. However, for trolleybuses that operate with catenary and perform dynamic recharging, consumption may vary:

1. **Standard Trolleybus (12m):** Consumption is in the range of 1.0 to 1.2 kWh/km under nominal load conditions.
2. **Articulated Trolleybus (18m):** Equipped with 151 kWh batteries, consumption increases due to higher mass and passenger capacity (up to 160 people), placing it in a range of 1.5 to 1.8 kWh/km.

Electrical Energy Tariffs and Supply

The STE receives energy through a network of 40 strategically distributed electrical substations, of which 39 have a capacity of 2,000 kW and one of 1,000 kW. The cost of this energy is subject to the CFE's Basic Supply tariffs for Large Demand in Medium Voltage (GDMTO and GDMTH).

For 2024 and 2025 projections, distribution tariffs in Mexico City have been adjusted according to inflation and generation costs. Regulated tariffs for August 2025 show variable energy charges ranging from 0.46 MXN/kWh in base periods to 1.22 MXN/kWh in peak periods for the GDMTH category. In addition, capacity and distribution charges that can exceed 200 MXN per kW of maximum demand must be considered. The weighted average price that STE pays for trolleybus operation has historically been estimated at 0.15 USD/kWh, including power charges.

Trolleybus Model	Battery Capacity (kWh)	Estimated Consumption (kWh/km)	Passenger Capacity
Yutong 12m	127	1.0 - 1.2	85
Yutong 18m	151	1.5 - 1.8	120 - 160

Detailed Operation of Line 2: Pantitlán - Chapultepec

Line 2 is a critical artery that crosses the city from east to west. Its strategic importance increased significantly after the fire at the Metro's Central Control Post in 2021, which led to an extension of its route to connect directly with CETRAM Pantitlán, adding 13 new stations to the original route.

Operational and Service Indicators

The line has a total length of 18.0 km and 34 stops. It serves the Cuauhtémoc, Iztacalco, and Venustiano Carranza boroughs, operating from 05:00 to 00:00 hours every day of the year.

For calculating densities in a mathematical model, the frequency of passage must be considered. Scheduled departures occur approximately every 8 minutes, implying high utilization of the catenary infrastructure and constant rotation of the rolling stock. In 2024, the trolleybus system as a whole mobilized over 90.7 million passengers, consolidating itself as the fourth most used electric transport in the capital. Line 2, being a high-traffic corridor, absorbs a significant portion of this demand, especially at interconnections with Metro Lines 1, 2, 3, 4, 5, 8, and 9.

Mileage and Fleet Performance

Annual mileage traveled is a fundamental variable for determining the asset replacement rate. Considering an operation of 19 hours a day with an average frequency of 8 minutes, approximately 142 dispatches are made per direction daily.

- **Daily kilometers per line:** .
- **Estimated annual kilometers:** .

This operational intensity justifies the investment in preventive maintenance, which for the STE implies carrying out more than 325,000 scheduled maintenance actions annually for the entire trolleybus network.

Parameters for the Line Integration Mathematical Model

To feed an optimization model based on densities, it is necessary to consolidate financial and technical data into precise input variables. The average values for the 2023-2025 period are presented below.

Direct Cost Variables

The cost function of the model must integrate the marginal cost per additional kilometer traveled and the fixed cost of infrastructure maintenance.

1. **Energy Cost:**

Taking a consumption of 1.1 kWh/km and an average tariff of 2.80 MXN/kWh (adjusted to 2024), the energy cost is approximately **3.08 MXN/km**.

2. **Unit Maintenance Cost:** Based on the budget of program E042 and the total mileage of the network, the cost of materials and services for maintenance is estimated at **2.15 MXN/km**.
3. **Personnel Cost:** Dividing the salary allocation of program E042 by the kilometers traveled by the total fleet, the human operation cost is around **12.50 MXN/km**.

Penalty and Surcharge Factors

The model must apply multipliers based on the geographical characteristics of the line segments:

- **Section with Tight Curves:** 20% increase in the maintenance cost of tires and carbon brushes.
- **Section with Elevated Structure:** 0.5% annual increase in the infrastructure value in the fixed maintenance cost.
- **Battery Use in Sections Without Catenary:** The operational cost increases slightly due to the chemical degradation of the LFP battery, estimated at a useful life of 8 years with a minimum SOC (State of Charge) of 10%.
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Input Parameter	Suggested Value (2024-2025)	Data Source
L2 Route Length	18.0 km (per direction)	
User Fare	\$4.00 MXN	

Input Parameter	Suggested Value (2024-2025)	Data Source
Energy Consumption (12m)	1.1 kWh/km	
Energy Consumption (18m)	1.8 kWh/km	
Unit Useful Life	20 years	
Maintenance Factor	+10% vs Diesel Bus	
Passengers per km (L2)	High Density (Transversal)	

Conclusions on Financial and Operational Sustainability

The analysis of Trolleybus Line 2 reveals a system with high energy efficiency but with a cost structure that critically depends on the preventive maintenance of the overhead electrical infrastructure. The transition to Yutong technology has allowed for a reduction in energy costs per passenger-kilometer, especially through the utilization of regenerative braking and autonomous operation capability.

From a financial perspective, STE's OPEX is heavily subsidized, as demonstrated by the gap between the 4 pesos fare and the actual operating costs derived from the Public Account. For an optimization model, this implies that passenger density is the variable that most influences the line's economic viability. The penalty for technical complexity on Eje 2 Sur is moderate compared to mountain areas, but the wear from electrical arcing on the catenary remains the main challenge for cost reduction in Chapter 2000 (Materiales y Suministros).

Finally, the surcharge for elevated infrastructure, observed in comparable CDMX systems, suggests that the integration of Line 2 with future viaducts or elevated passes will require a budgetary increase proportional to the logistical complexity of maintenance. The use of real data from 2023 and 2024 allows for the projection that Line 2 will continue to be the reference corridor for surface electrical efficiency, provided that investment in catenary adjustment and timely renewal of contact components is maintained.

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